15. Draw a triangle ABC of base $BC = 5.6$ cm, $\angle A = 40°$ and the bisector of $\angle A$ meets BC at D such that $CD = 4$ cm.

16. Draw ΔPQR such that $PQ = 6.8$ cm, vertical angle is $50°$ and the bisector of the vertical angle meets the base at D where $PD = 5.2$ cm.

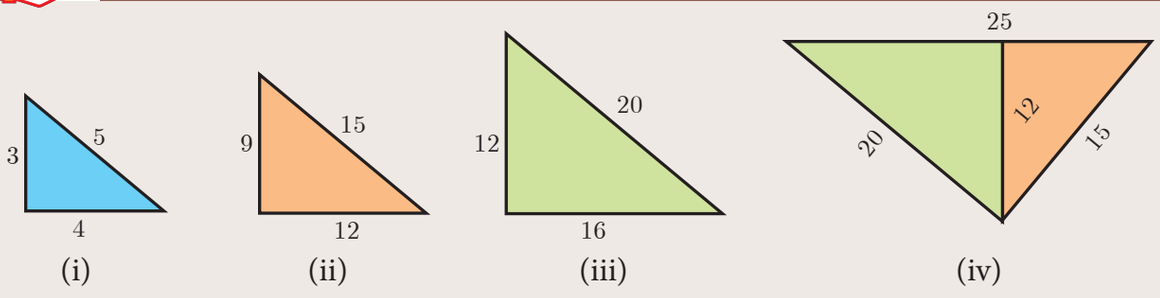
4.4 Pythagoras Theorem

Among all existing theorems in mathematics, Pythagoras theorem is considered to be the most important because it has maximum number of proofs. There are more than 350 ways of proving Pythagoras theorem through different methods. Each of these proofs was discovered by eminent mathematicians, scholars, engineers and math enthusiasts, including one by the 20th American president James Garfield. The book titled "The Pythagorean Proposition" written by Elisha Scott Loomis, published by the National Council of Teaching of Mathematics (NCTM) in America contains 367 proofs of Pythagoras Theorem.

Three numbers (a, b, c) are said to form Pythagorean Triplet, if they form sides of a right triangle. Thus (a, b, c) is a Pythagorean Triplet if and only if $c^2 = a^2 + b^2$.

Now we are in a position to study this most famous and important theorem not only in Geometry but in whole of mathematics.

Activity 4



(i) (ii) (iii) (iv)

Fig. 4.45

Step 1: Take a chart paper, cut out a right angled triangle of measurement as given in triangle (i) .

Step 2: Take three more different colour chart papers and cut out three triangles such that the sides of triangle (ii) is three times of the triangle (i), the sides of triangle (iii) is four times of the triangle (i), the sides of triangle (iv) is five times of triangle (i).

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